Photosynthesis

Photosynthesis is the process plants use to make their own food. Instead of eating, a plant captures energy from sunlight and stores it in sugar. Almost all life on Earth depends on this process, directly or indirectly.

Where it happens

Photosynthesis takes place mostly in the leaves. Inside the cells of a leaf are small structures called chloroplasts. Chloroplasts contain a green pigment called chlorophyll, which absorbs sunlight. The green colour of leaves comes from chlorophyll.

What a plant needs

A plant needs three things for photosynthesis. The first is sunlight, which provides the energy. The second is water, which the roots absorb from the soil and send up to the leaves. The third is carbon dioxide, a gas that enters the leaf through tiny openings called stomata.

What a plant produces

The plant uses the energy from sunlight to turn water and carbon dioxide into glucose, a sugar, and oxygen. The plant uses glucose for growth and energy, or stores it as starch. The oxygen is released into the air through the stomata. Most of the oxygen we breathe was produced by photosynthesis.

The chemical equation

The process can be summarised as: 6CO2 + 6H2O + light energy -> C6H12O6 + 6O2. Six molecules of carbon dioxide and six molecules of water produce one molecule of glucose and six molecules of oxygen.

Two stages

Photosynthesis has two stages. In the light-dependent reactions, chlorophyll captures light energy and uses it to split water, releasing oxygen. In the Calvin cycle, the energy captured in

the first stage is used to build glucose from carbon dioxide. The Calvin cycle does not need light directly, but it depends on the products of the light-dependent reactions.

What changes the rate

Three factors affect how fast photosynthesis happens: light intensity, the amount of carbon dioxide, and temperature. More light speeds up photosynthesis until another factor becomes the limit. Very high temperatures damage the enzymes involved and slow the process down.

Why it matters

Photosynthesis provides the food at the start of nearly every food chain, and it supplies the oxygen that animals need. It also removes carbon dioxide from the air, which is one reason forests matter for the climate.